HS4002: Lecture 4B

Bivariate Statistics

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Correlations between Continuous Variables

Pearson's Correlation Coefficient

- Pearson's correlation coefficient measures the standardized covariance between <u>two continuous variables</u>.
- It measures the linear association between the two continuous variables.
- It is the ratio between the covariance of two variables and the product of their standard deviations.

Pearson's Correlation Coefficient

$$r_{xy} = \frac{\mathrm{cov}_{xy}}{s_x s_y}$$

$$= \frac{(x_i - \bar{x})(y_i - \bar{y})}{(N-1)\sqrt{\frac{1}{N-1}(x_i - \bar{x})^2}\sqrt{\frac{1}{N-1}(y_i - \bar{y})^2}}$$

Interpreting Pearson's r

- Pearson's r can span between -1 and +1.
  - → +1 implies perfect positive linear correlation.
  - → -1 implies perfect negative linear correlation.
  - → 0 implies no linear relationship.
- Hypothesis testing:
  - → r has a known t-distribution which we can use to produce a p-value.

Correlations between Ordinal Variables

Spearman's Correlation Coefficient

- Spearman's correlation coefficient measures the rank correlation between two variables.
  - $\rightarrow$ It is appropriate for both ordinal and continuous variables.
- It measures the monotonicity of the relationship between two variables.

Spearman's Correlation Coefficient

After the raw values of X and Y are converted to ranks $R(X)$ and $R(Y)$,

$$\rho_{xy} = \frac{\mathrm{cov}(R(X), R(Y))}{\sigma_{R(X)}\sigma_{R(Y)}}$$

Interpreting Spearman's ρ

- Spearman's ρ can span between -1 and +1.
  - $\rightarrow$ +1 implies perfectly monotonically increasing relationship (when X increases, Y always increases).
  - $\rightarrow$ -1 implies perfectly monotonically decreasing relationship (when X increases, Y always decreases).
  - $\rightarrow$ 0 implies no monotonic relationship (there is no tendency for Y to either increase or decrease when X increases).
- Hypothesis testing:
  - $\rightarrow$ ρ has a known t-distribution which we can use to produce a p-value.

Kendall's Rank Correlation Coefficient

- Kendall's rank correlation coefficient is an alternative measure of the rank correlation between two variables.
  - $\rightarrow$ It is also appropriate for both ordinal and continuous variables.
- It is better for instances where there are many scores with the same rank.

Kendall's Rank Correlation Coefficient

$$\tau_{xy} = \frac{\text{no. of concordant pairs - no. of discordant pairs}}{\text{no. of pairs}}$$

Interpreting Kendall's τ

- Kendall's τ can span between -1 and +1.
  - $\rightarrow$ +1 implies perfect agreement/concordance among the observed ranks of the two variables.
  - $\rightarrow$ -1 implies implies perfect disagreement/discordance among the observed ranks of the two variables.
  - $\rightarrow$ 0 implies the ranks of the two variables are independent of each other.
- Hypothesis testing:
  - $\rightarrow$ τ has an approximately normal distribution when n is large.

Correlations between Nominal Variables

Cramer's V

- Kramer's V is a measure of association between two nominal variables.
- It is based on the Chi-squared test.

Cramer's V

$$V_{xy} = \sqrt{\frac{\chi^2/n}{\min(k-1, r-1)}}$$

where k is the no. of dimensions of X, and r is the number of dimensions of Y, χ^2 is from the Pearson's chi-squared statistic, n is the total number of observations.

Interpreting Cramer's V

- Cramer's V can span between 0 and $+1$.
  - $\rightarrow$ $+1$ implies that one variable is completely determined by the other.
  - $\rightarrow$ 0 implies no association between the variables.
- Hypothesis testing:
  - $\rightarrow$ V follows a Chi-squared distribution.

Example: Lieberson & Bell (1992)'s work on social taste.

Lieberson & Bell (1992)

- Lieberson and Bell use naming practices as a case of pure taste, free of organized efforts to influence taste.
- More specifically, they look at the names given to children in NYC from 1973 to 1985.

Lieberson & Bell (1992)

- Two categorical variables:
  - $\rightarrow$ Top 20 most popular names.
  - $\rightarrow$ Year.
- What would Cramer's V tell you?
  - $\rightarrow$ What does a high Cramer's V between the two tell you?
  - $\rightarrow$ What about a low Cramer's V?

Gendered Differences

Figure 1: Gendered differences in change of naming tastes

The results are very clear: there is less stability and more turnover in the top 20 names given to girls during the span under consideration; V is .40 and .24, respectively, for girls' and boys' names. These "standardized" chi-square values indicate that the most popular names for girls are more closely associated with year; that is, they change more between 1973 and 1985 than do boys' names. The disposition toward new names

There is more turnover in the change of girl's names from 1973 to 1985.

Figure 2: Differences by mother's education

TABLE 6

EDUCATIONAL DIFFERENCES IN AGGREGATE NAMING CHARACTERISTICS

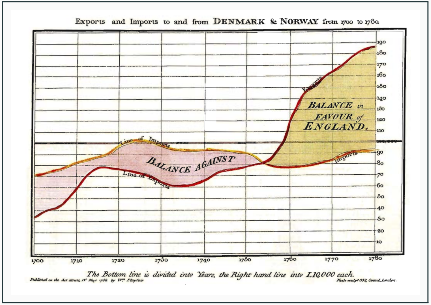
Years of Education of Mother	% in Top 20	Index of Unique Names	Cramer's V, Leading Names, 1973–85
Mothers of girls:			
9–11	27.8	48.0	.56
12	33.4	36.7	.49
13–15	35.6	36.6	.43
16	36.3	35.2	.39
17+	35.6	34.9	
Mothers of boys:			
9–11	40.7	35.8	.33
12	48.0	25.7	.28
13–15	49.6	21.8	.29
16	49.6	23.2	.24
17+	48.8	17.8	

NOTE.—All values of Cramer's V are based on chi-square significance at .001.

For both girls and boys: turnover in names is higher among the less-educated.

Data Visualization for Bivariate Statistics

Figure 3: Playfair's graph of British trade



Data visualization refers to the visual display of measured quantities by means of the combined use of points, lines, a coordinate system, numbers, symbols, words, shading and color.

How Visualization Helps

- Good visualization facilitates inspection; they make large data sets coherent.
- They are narratives of data, the metaphorical "thousand words."

Figure 4: Anscombe's Quartet

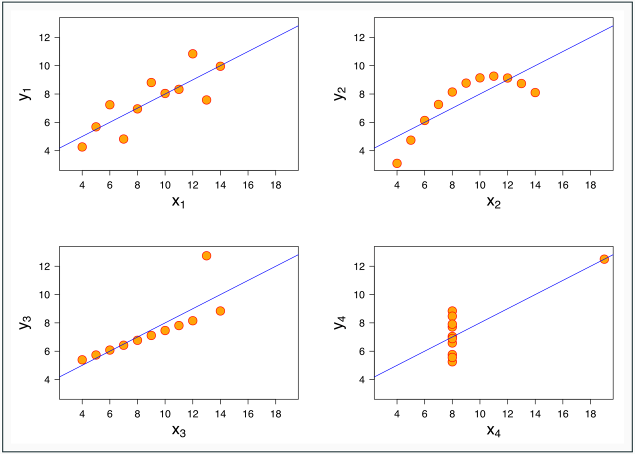
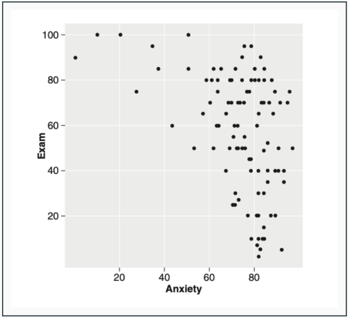


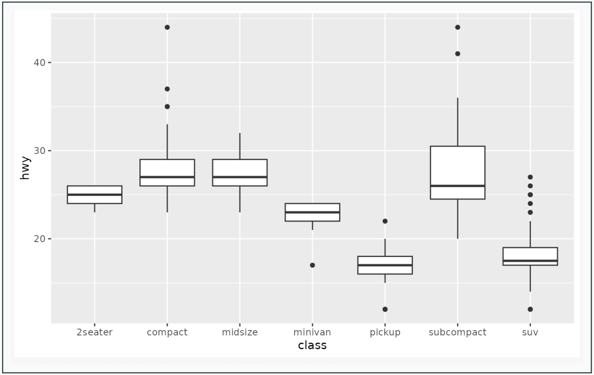
Figure 5: An Example of a Scatterplot



We use scatterplots to visualize the relationship between two continuous variables.

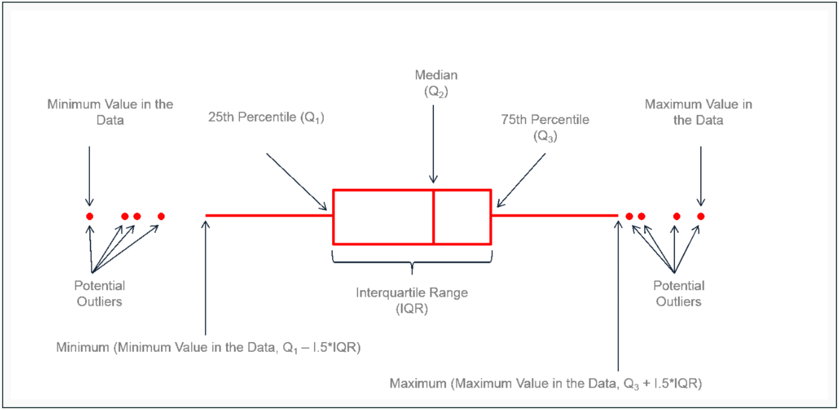
Boxplots

Figure 6: An Example of a Boxplot



We use boxplots to visualize the relationship between a categorial variable and a continuous variable.

Figure 7: Anatomy of a Boxplot



Thanks for listening :)